How vaccines train the immune system

A vaccine shows your immune system a harmless piece or a weakened version of a germ, so it can practise before it meets the real thing. The unfamiliar piece is called an antigen.

When the body sees the antigen, white blood cells make antibodies that can latch onto it. Some of those cells stay behind as memory cells. If the real germ shows up months or years later, the memory cells recognise it and the body responds fast, often before you feel sick.

Vaccines come in several types. Some use a weakened live germ, some use a killed germ, some use only a protein from the germ's surface, and newer mRNA vaccines give your cells instructions to make that protein themselves for a short time.

Booster doses remind the immune system about the antigen, because protection from some vaccines fades over time. Mild side effects such as a sore arm or a day of tiredness are a sign the immune system is responding.